Problem 1

1. $\{\emptyset\}$
2. $\emptyset$
3. $\{\{\emptyset\}, \emptyset\}$
4. $\emptyset$

Problem 2

Claim: For all sets A, B, C, and D, if $A \subseteq B$ and $C \subseteq D$, then $(A \cap C) \subseteq (B \cap D)$.

Let A, B and C be arbitrary sets where $A \subseteq B$ and $C \subseteq D$
We will prove that $(A \cap C) \subseteq (B \cap D)$. Let us choose an arbitrary $x \in (A \cap C)$ then $x \in A$ and $x \in C$.

- Since $A \subseteq B$, and we know $x \in A$, it follows that $x \in B$

Since $C \subseteq D$ and we know $x \in C$, it follows that $x \in D$

Now we know that $x \in B$ and $x \in D$ then $x \in (B \cap D)$.

Since we already showed that $x \in (A \cap C)$ and showed that x must also be in $(B \cap D)$ then $(A \cap C) \subseteq (B \cap D)$

Problem 3

Claim: If A and B are any sets, then
$P(A) \cap P(B) = P(A \cap B)$

To show that they are equal we must also prove that
$P(A \cap B) \subseteq P(A) \cap P(B)$

Let A and B be sets. Consider an arbitrary set $S \in P(A \cap B)$. This means we need to prove $S \subseteq A \cap B$.

Since $S \subseteq (A \cap B)$ and $(A \cap B) \subseteq A$, it follows that $S \subseteq A$

Since $S \subseteq (A \cap B)$ and $(A \cap B) \subseteq B$, this means $S \subseteq B$

Because $S \subseteq A$ then $S \in P(A)$ and since $S \subseteq B$ then $S \in P(B)$

Since $S \in P(A)$ and $S \in P(B)$, then $S \in P(A) \cap P(B)$.

Thus we have shown that $P(A \cap B) \subseteq P(A) \cap P(B)$